

STRUCTURA: A Bayesian Network Framework for Information-Theoretic Scoring and Behavioral Anomaly Detection

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01 · INTRODUCTION

Under network psychometrics, constructs are modeled as interacting nodes linked by explicit dependencies. Bayesian Networks enable exact person-level inference^{1,2}, but their use in cognitive assessment remains limited. **No validated framework currently turns a single response pattern into a reliable, interpretable, and clinically meaningful score³.** STRUCTURA would address this gap.

03 · RESULTS – THE GUTTMAN SCALE PARADOX

A Guttman paradox⁷: failing easy items while solving difficult ones.

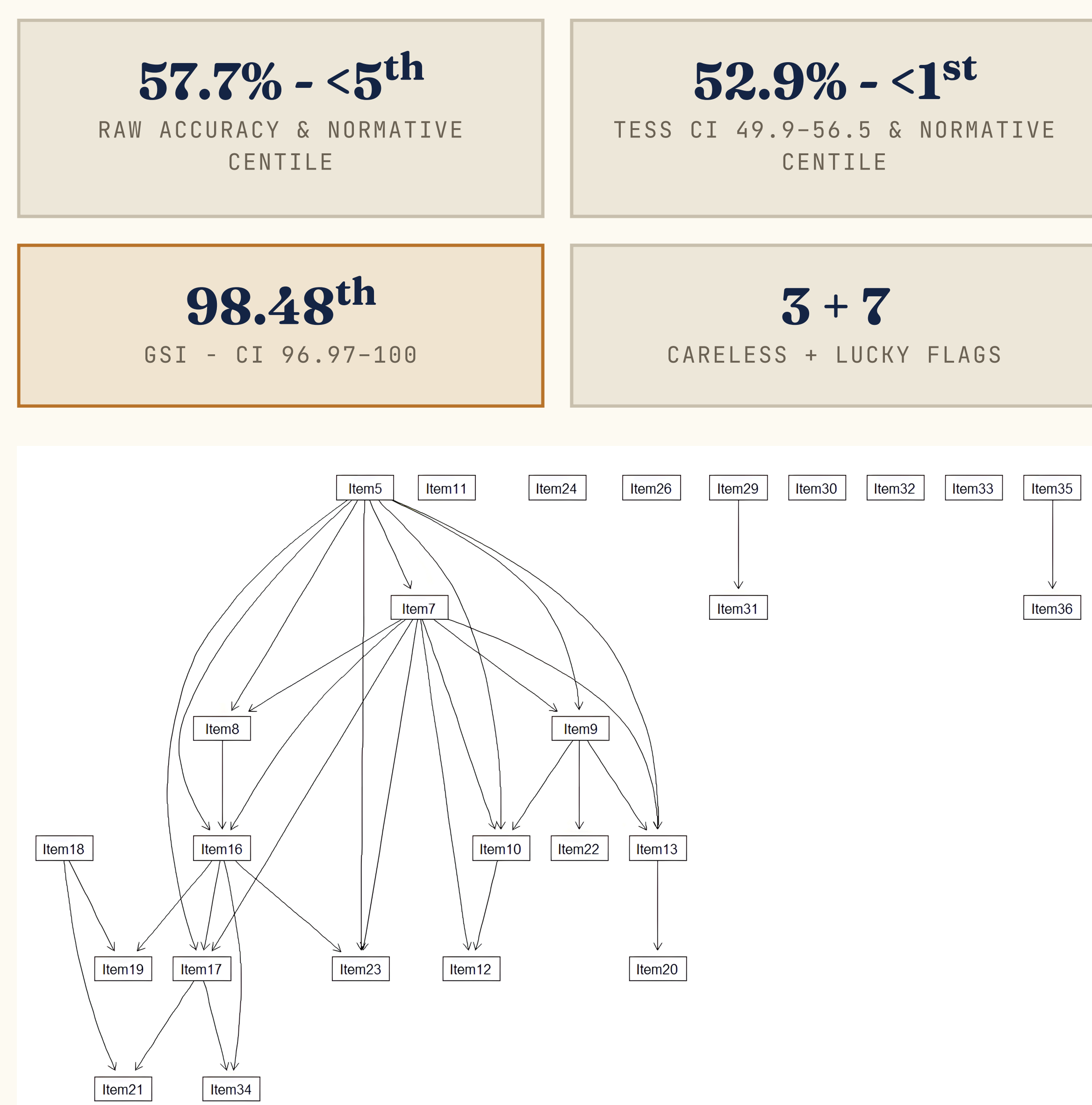


Fig. 1 — Learned normative BN of Raven's CPM, adapted from Orsoni et al. (2025)⁵.

TESS Difference (Raw – TESS = +4.76) is the structural discrepancy.

GSI is more structurally surprising than 98.48% of the normative sample.

LSI the framework flags individual response anomalies (Fig.2).

02 · METHODS

Normative sample & instrument. Normative data: 32 kindergarten children ($M_{age} = 4.72$, $SD = 0.30$; 46.9% male) from Orsoni et al. (2025). Raven's CPM, a non-verbal measure of fluid intelligence and analogical reasoning, includes 36 dichotomous items across subscales A, Ab, and B; after excluding zero-variance items, 26 items were analysed.

2 Compilation & local evidence. The BN was compiled into a junction tree for exact inference, with Laplace-smoothed CPTs ($\alpha = 0.5$). For each item, local evidence was defined by its Markov Blanket: $E_i = MB(X_i)$. Only structurally relevant neighbours informed person-level inference.

3 TESS. $\sum_i P(X_i = 1 | E_i)$, normalised to 0–100. Equal raw accuracy can yield different TESS values.

4 Global SI (GSI). $\sum_i -\log_2 P(X_i = x_i | E_i)$, mapped onto an empirical 0–100 normative centile scale.

5 Local SI (LSI). $SI_{\pm, i} = (x_i - E[X_i | E_i]) \cdot I(x_i)$: signed item-level anomaly; flagged at the 5th/95th normative percentiles.

6 Uncertainty. Parametric bootstrap (95% CI): each CPT perturbed via Dirichlet sampling (ISS as hyperparameter).

04 · DISCUSSION & CONCLUSIONS

These three metrics capture complementary aspects of anomaly: **TESS** adjusts the ability estimate for structural inconsistency, **GSI** quantifies how improbable the overall response pattern is, and **LSI** identifies the specific items driving the anomaly.

STRUCTURA translates BN topology into individual-level scoring.

Uncertainty is rigorously quantified via Dirichlet bootstrap CIs.

Generalizable to any assessment where a DAG is theoretically or empirically justified.

Next steps: Formally validate TESS, GSI, and LSI; integrating contextual nodes and evaluating clinical feasibility in psychiatric populations.

03B · RESULTS – LOCAL SI PROFILE (GUTTMAN CASE)

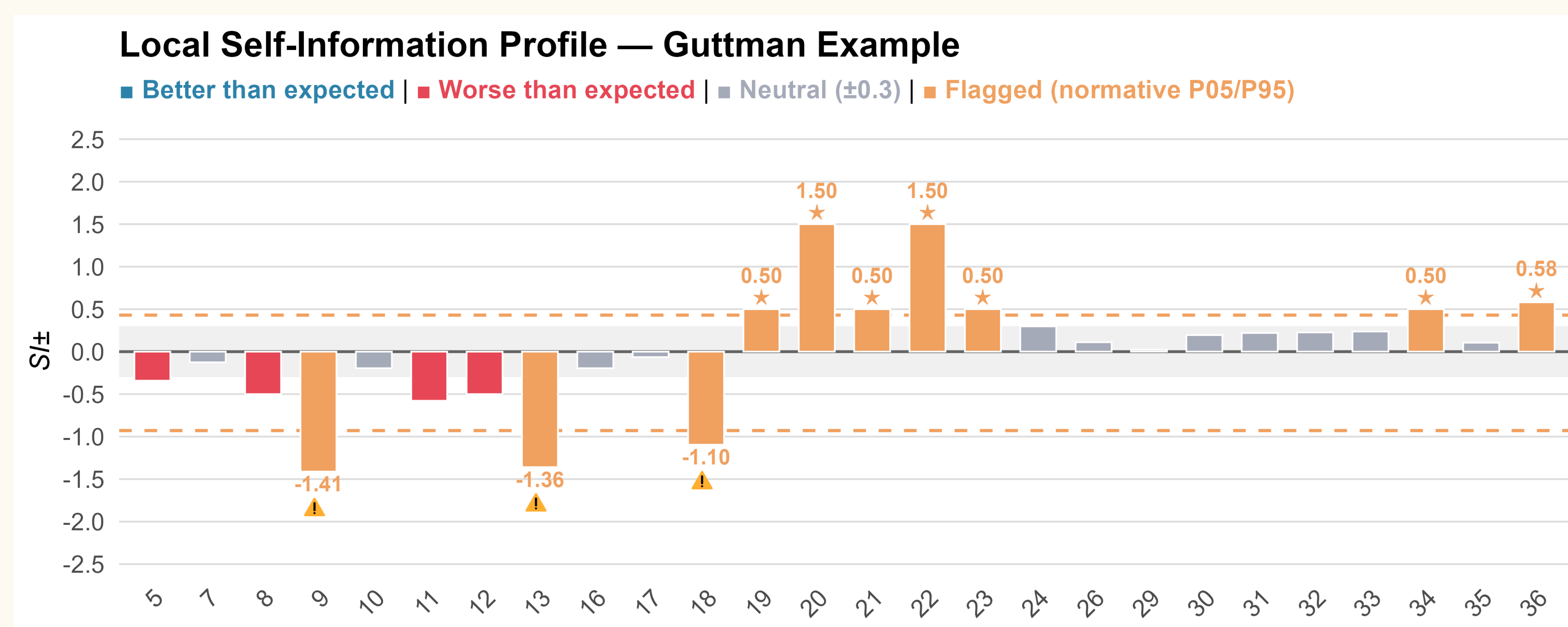


Fig. 2 — Signed SI per item; flags at normative P05 = -0.929 and P95 = 0.430.